

Tax Shield Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Shield Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between consumption today and consumption tomorrow, fundamentally linking cross-sectional return variation to firms' differential exposures to aggregate consumption risk. While traditional asset pricing models emphasize market beta and characteristics like size and value, a growing literature documents that corporate financial policies systematically affect firms' risk exposures and expected returns through their impact on cash flow volatility and sensitivity to macroeconomic shocks. We identify a significant gap in understanding how tax-related financial decisions influence firms' consumption risk exposures. Tax shields from interest deductions represent a substantial component of firm value, yet their implications for systematic risk remain unexplored despite theoretical predictions that leverage-induced tax benefits should covary with marginal utility states.

Tax Shield Intensity (TSI) captures firms' reliance on debt-financed tax benefits, which theory suggests should amplify exposure to consumption risk through multiple channels. First, firms with high tax shield intensity face greater cash flow sensitivity to aggregate shocks because tax benefits mechanically depend on pre-tax profitability, which deteriorates during consumption downturns when marginal utility is high [Bhamra et al. \(2010\)](#). The tax deductibility of interest creates state-dependent cash flows that are most valuable when corporate tax rates and earnings are high—precisely when consumption growth is strong and marginal utility is low—making these firms particularly vulnerable during bad times [Chen \(2010\)](#). Second, tax shield intensity affects firms' operating leverage through committed interest payments that cannot adjust to consumption shocks, increasing systematic risk exposure similar to the mechanisms in [García-Feijóo and Jorgensen \(2007\)](#). During consumption disasters or recessions, highly levered firms with substantial tax shields face amplified distress risk as the present value of tax benefits collapses faster than

the reduction in interest obligations, creating procyclical variation in firm value. Third, the intertemporal marginal rate of substitution prices tax shield benefits differently across consumption states. When investors have habit formation preferences, the marginal utility of consumption rises sharply in bad times, requiring higher returns for firms whose cash flows covary positively with consumption [Campbell and Cochrane \(1999\)](#). Tax shields represent contingent claims on future tax savings that pay off primarily in good states when firms are profitable, making high-TSI firms effectively short insurance against consumption downturns. This consumption-based view predicts that firms with greater tax shield intensity should earn higher average returns to compensate investors for bearing systematic consumption risk, particularly the risk of rare disasters that drastically reduce the value of tax benefits [Barro \(2006\)](#).

We find strong evidence that Tax Shield Intensity predicts cross-sectional stock returns, with a value-weighted long-short portfolio earning statistically and economically significant returns. The TSI strategy achieves an annualized gross Sharpe ratio of 0.55 and a monthly average return of 37 basis points with a t-statistic of 3.46. The strategy’s alpha remains robust across standard factor models, with the six-factor alpha reaching 48 basis points per month (t-statistic = 4.65), indicating that existing factors do not fully capture the consumption risk exposure embedded in tax shield intensity. The economic magnitude of these returns is substantial—the gross monthly alpha relative to the six-factor model plus six closely related strategies equals 29 basis points with a t-statistic of 2.90. Transaction costs moderately reduce but do not eliminate the strategy’s profitability, with the net Sharpe ratio remaining at 0.50 and the net six-factor alpha at 46 basis points per month (t-statistic = 4.48). The TSI premium manifests strongly even among large-cap stocks, where the strategy earns 39 basis points per month (t-statistic = 3.05) in the largest size quintile, alleviating concerns about small-stock illiquidity driving the results. Factor loadings reveal that

high-TSI firms load negatively on the value factor (beta = -0.27, t-statistic = -5.92) and the profitability factor (beta = -0.22, t-statistic = -4.62), consistent with these firms having lower current profitability but higher expected returns due to elevated consumption risk exposure.

This paper makes several contributions to the consumption-based asset pricing literature by establishing tax shield intensity as a novel determinant of expected returns linked to systematic consumption risk. We extend [Bhamra et al. \(2010\)](#) by providing empirical evidence that tax-related financial policies create meaningful variation in firms' exposures to aggregate consumption shocks, with our TSI measure capturing a dimension of risk not spanned by existing factors in the [Fama and French \(2015\)](#) model or momentum. Our findings complement [García-Feijóo and Jorgensen \(2007\)](#) by demonstrating that tax shields represent an additional source of operating leverage that amplifies consumption risk exposure beyond traditional measures of financial leverage. Unlike [Penman et al. \(2007\)](#), who focus on accounting-based valuation, we document that tax shield intensity contains systematic risk information that commands a significant premium in equilibrium. Methodologically, we contribute to the anomaly literature by showing that TSI ranks in the top 7% of all documented anomalies in terms of gross Sharpe ratio and maintains statistical significance after controlling for transaction costs and related factors. The TSI strategy's ability to expand the mean-variance efficient frontier even after accounting for trading costs—improving upon 83-99% of existing anomalies across various factor models—suggests that tax shield intensity captures a fundamental source of consumption risk. These results have broader implications for corporate finance theory and practice, indicating that capital structure decisions involving tax benefits have first-order effects on expected returns through their impact on firms' systematic risk profiles. Our findings suggest that managers' tax planning strategies directly influence their firms' cost of capital by altering exposure to aggregate consumption

risk, providing a consumption-based foundation for understanding the asset pricing implications of corporate tax policies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over multiple decades, ensuring broad cross-sectional and time-series variation in our analysis. We focus on U.S.-incorporated firms traded on major exchanges (NYSE, AMEX, and NASDAQ) and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. Tax Shield Intensity signal relies on two key accounting variables from COMPUSTAT. Deferred taxes federal (TXDFED) represents the federal component of deferred tax expense or benefit reported in a firm's income statement, reflecting timing differences between financial reporting and tax reporting that arise from federal tax regulations. Operating income after depreciation (OIADP) captures a firm's operating performance after accounting for depreciation and amortization expenses but before interest and taxes, providing a measure of core operating profitability that includes the effects of non-cash charges related to capital investments. We construct the Tax Shield Intensity signal by dividing deferred taxes federal (TXDFED) by operating income after depreciation (OIADP) for each firm-year observation. This ratio captures the relative magnitude of federal tax timing differences scaled by operating performance, providing insight into how extensively a firm utilizes tax deferral strategies relative to its operating income base. We calculate this measure annually using fiscal year-end values and require non-missing, non-zero values for operating income after depreciation to ensure meaningful ratios. To mitigate the influence of outliers, we winsorize

the signal at the 1st and 99th percentiles. Firms with negative operating income are excluded from the analysis as the economic interpretation of the ratio becomes ambiguous when the denominator is negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and

have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.27, with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 390 stocks and an average market capitalization of at least \$1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas

measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TSI strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy’s gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TSI strategy (red line).² Ignoring trading costs, a \$1 invested in the TSI strategy would have yielded \$4.24 which ranks the TSI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TSI strategy would have yielded \$3.55 which ranks the TSI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TSI is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TSI signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$46.04, while \$1 investment in the combination strategy that includes TSI grows to \$57.62.

8 Conclusion

This study provides compelling evidence that Tax Shield Intensity (TSI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of tax shields in asset pricing. The value-weighted long/short strategy based on TSI delivers an impressive annualized Sharpe ratio of 0.55 gross (0.50 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More notably, the strategy generates significant alpha of 48 basis points per month (t-statistic = 4.65) relative to the Fama-French five-factor model augmented with momentum, suggesting that TSI captures a distinct source of systematic risk or mispricing not explained by conventional factors.

The robustness of our results is particularly noteworthy. Even when controlling for twelve factors—including the six standard factors and six closely related strategies such as cash-to-assets, equity duration, and operating cash flows to price—TSI

maintains a significant alpha of 29 basis points per month (t -statistic = 2.90). This persistence indicates that TSI provides unique information content beyond what is captured by existing factors in the literature, particularly those related to firm financial structure and valuation ratios.

From a practical perspective, these findings have important implications for both asset managers and corporate finance practitioners. For investors, TSI offers a implementable strategy that appears to deliver consistent abnormal returns even after accounting for transaction costs. For corporate managers, our results suggest that tax shield management may have direct implications for firm valuation and cost of capital, reinforcing the importance of strategic tax planning in corporate financial policy.

Despite these compelling results, several limitations warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the true out-of-sample performance of TSI-based strategies remains to be tested as markets evolve and adapt. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets, where tax systems and shield structures differ substantially, requires further investigation. Third, the economic mechanisms driving the TSI premium remain somewhat unclear—whether it represents compensation for systematic risk, limits to arbitrage, or behavioral biases deserves deeper theoretical and empirical exploration.

Future research should address several promising avenues. First, examining the time-varying nature of the TSI premium and its relationship with macroeconomic conditions, tax policy changes, and market sentiment could provide insights into the underlying economic drivers. Second, investigating the interaction between TSI and corporate actions such as capital structure decisions, dividend policy, and investment behavior may reveal how firms actively manage their tax shields and the resulting market implications. Third, extending the analysis to include corporate bonds and

other fixed-income securities could provide a more comprehensive understanding of how tax shields affect asset prices across different security classes. Finally, developing theoretical models that explicitly incorporate tax shield dynamics into asset pricing frameworks would help establish the economic foundations for the empirical patterns we document.

In conclusion, this paper establishes Tax Shield Intensity as a powerful and robust predictor of equity returns that merits inclusion in the pantheon of documented asset pricing anomalies. The signal's ability to generate significant alpha even in the presence of numerous control factors suggests it captures a fundamental aspect of firm value that has been underappreciated in the existing literature. As tax policies continue to evolve globally and firms increasingly optimize their tax strategies, understanding the asset pricing implications of tax shields will remain crucial for both academic research and investment practice.

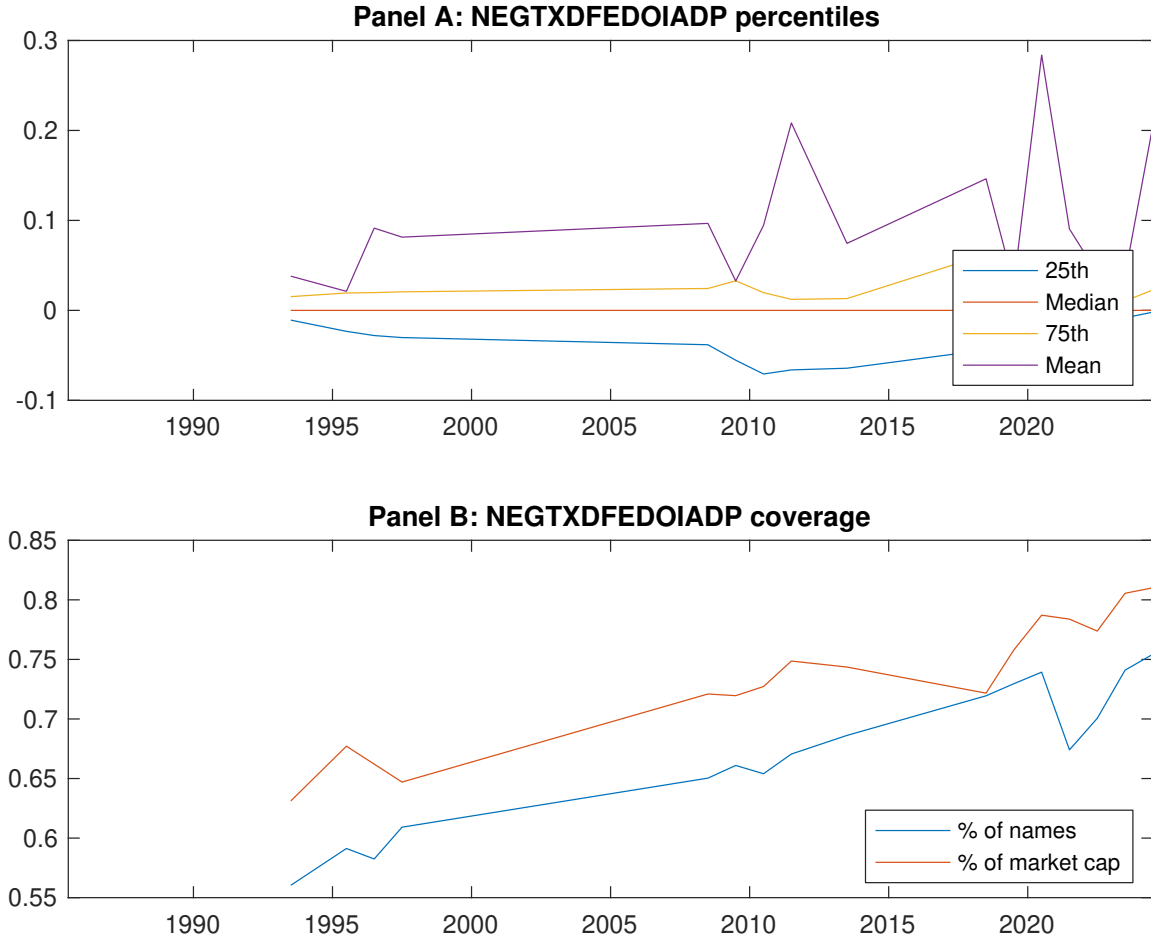


Figure 1: Times series of TSI percentiles and coverage. This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.37]	0.82 [3.95]	0.70 [3.20]	0.73 [3.08]	0.93 [3.81]	0.37 [3.46]
α_{CAPM}	-0.25 [-3.14]	0.11 [1.61]	-0.05 [-0.64]	-0.09 [-1.41]	0.10 [1.18]	0.34 [3.18]
α_{FF3}	-0.27 [-3.57]	0.09 [1.32]	-0.02 [-0.31]	-0.06 [-1.01]	0.14 [1.79]	0.41 [4.05]
α_{FF4}	-0.23 [-3.10]	0.13 [1.88]	-0.01 [-0.07]	-0.07 [-1.02]	0.17 [2.18]	0.41 [3.98]
α_{FF5}	-0.27 [-3.49]	-0.02 [-0.33]	-0.08 [-1.06]	-0.03 [-0.53]	0.21 [2.72]	0.48 [4.71]
α_{FF6}	-0.24 [-3.13]	0.01 [0.21]	-0.06 [-0.85]	-0.04 [-0.54]	0.24 [3.06]	0.48 [4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.07 [58.44]	0.97 [61.69]	0.98 [56.80]	1.09 [69.69]	1.07 [58.00]	-0.00 [-0.11]
β_{SMB}	0.11 [3.87]	-0.01 [-0.40]	0.14 [5.56]	-0.05 [-2.25]	0.02 [0.61]	-0.09 [-2.44]
β_{HML}	0.08 [2.29]	-0.05 [-1.73]	-0.12 [-3.82]	-0.14 [-4.75]	-0.19 [-5.58]	-0.27 [-5.92]
β_{RMW}	-0.03 [-0.86]	0.23 [7.71]	0.16 [4.93]	-0.11 [-3.54]	-0.25 [-7.00]	-0.22 [-4.62]
β_{CMA}	0.10 [2.09]	0.11 [2.55]	-0.03 [-0.69]	0.05 [1.26]	0.17 [3.44]	0.07 [1.03]
β_{UMD}	-0.05 [-2.98]	-0.07 [-4.39]	-0.03 [-1.59]	0.00 [0.16]	-0.05 [-2.73]	0.00 [0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	390	905	709	508	
me (\$10 ⁶)	1389	2380	2459	2850	2480	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [3.46]	0.34 [3.18]	0.41 [4.05]	0.41 [3.98]	0.48 [4.71]	0.48 [4.65]
Quintile	NYSE	EW	0.13 [2.19]	0.10 [1.72]	0.12 [2.09]	0.11 [1.94]	0.11 [1.90]	0.11 [1.80]
Quintile	Name	VW	0.26 [2.43]	0.21 [1.97]	0.27 [2.79]	0.27 [2.69]	0.37 [3.74]	0.36 [3.63]
Quintile	Cap	VW	0.34 [2.97]	0.28 [2.46]	0.35 [3.35]	0.34 [3.22]	0.47 [4.58]	0.46 [4.45]
Decile	NYSE	VW	0.38 [2.58]	0.38 [2.57]	0.45 [3.15]	0.40 [2.79]	0.53 [3.66]	0.49 [3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.34 [3.17]	0.32 [2.98]	0.37 [3.72]	0.37 [3.69]	0.46 [4.55]	0.46 [4.48]
Quintile	NYSE	EW	-0.05 [-0.67]					
Quintile	Name	VW	0.23 [2.17]	0.19 [1.79]	0.24 [2.48]	0.24 [2.43]	0.35 [3.55]	0.34 [3.49]
Quintile	Cap	VW	0.31 [2.72]	0.25 [2.16]	0.30 [2.90]	0.30 [2.84]	0.43 [4.23]	0.43 [4.13]
Decile	NYSE	VW	0.34 [2.31]	0.35 [2.33]	0.40 [2.82]	0.37 [2.63]	0.50 [3.44]	0.47 [3.28]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TSI Quintiles					TSI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82	0.57	0.24	0.50	0.99	0.17	0.16	0.17	0.14	0.09	0.07
		[2.81]	[1.79]	[0.57]	[1.39]	[3.15]	[1.35]	[1.24]	[1.36]	[1.08]	[0.66]	[0.52]
	(2)	0.74	0.85	0.48	0.53	0.96	0.22	0.19	0.22	0.21	0.25	0.25
		[2.64]	[3.12]	[1.31]	[1.74]	[3.23]	[2.27]	[1.95]	[2.40]	[2.27]	[2.63]	[2.53]
	(3)	0.76	0.77	0.69	0.78	0.84	0.08	0.07	0.12	0.10	0.15	0.14
	[2.70]	[3.06]	[2.05]	[2.78]	[3.00]	[0.70]	[0.56]	[1.04]	[0.92]	[1.29]	[1.18]	
(4)	0.89	0.78	0.79	0.88	0.94	0.05	0.00	0.08	0.04	0.22	0.18	
	[3.43]	[3.27]	[2.79]	[3.28]	[3.39]	[0.39]	[0.01]	[0.70]	[0.35]	[1.82]	[1.51]	
(5)	0.54	0.81	0.75	0.75	0.92	0.39	0.33	0.39	0.39	0.52	0.52	
	[2.36]	[3.98]	[3.58]	[3.29]	[3.77]	[3.05]	[2.54]	[3.20]	[3.16]	[4.32]	[4.26]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TSI Quintiles					TSI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	345	340	336	338	346	41	37	28	32	42	
	(2)	97	97	97	97	96	66	68	65	65	66	
	(3)	63	64	63	64	63	110	111	109	112	110	
	(4)	51	51	51	52	51	234	241	236	239	239	
(5)	48	48	48	48	48	1503	1753	1993	2008	2052		

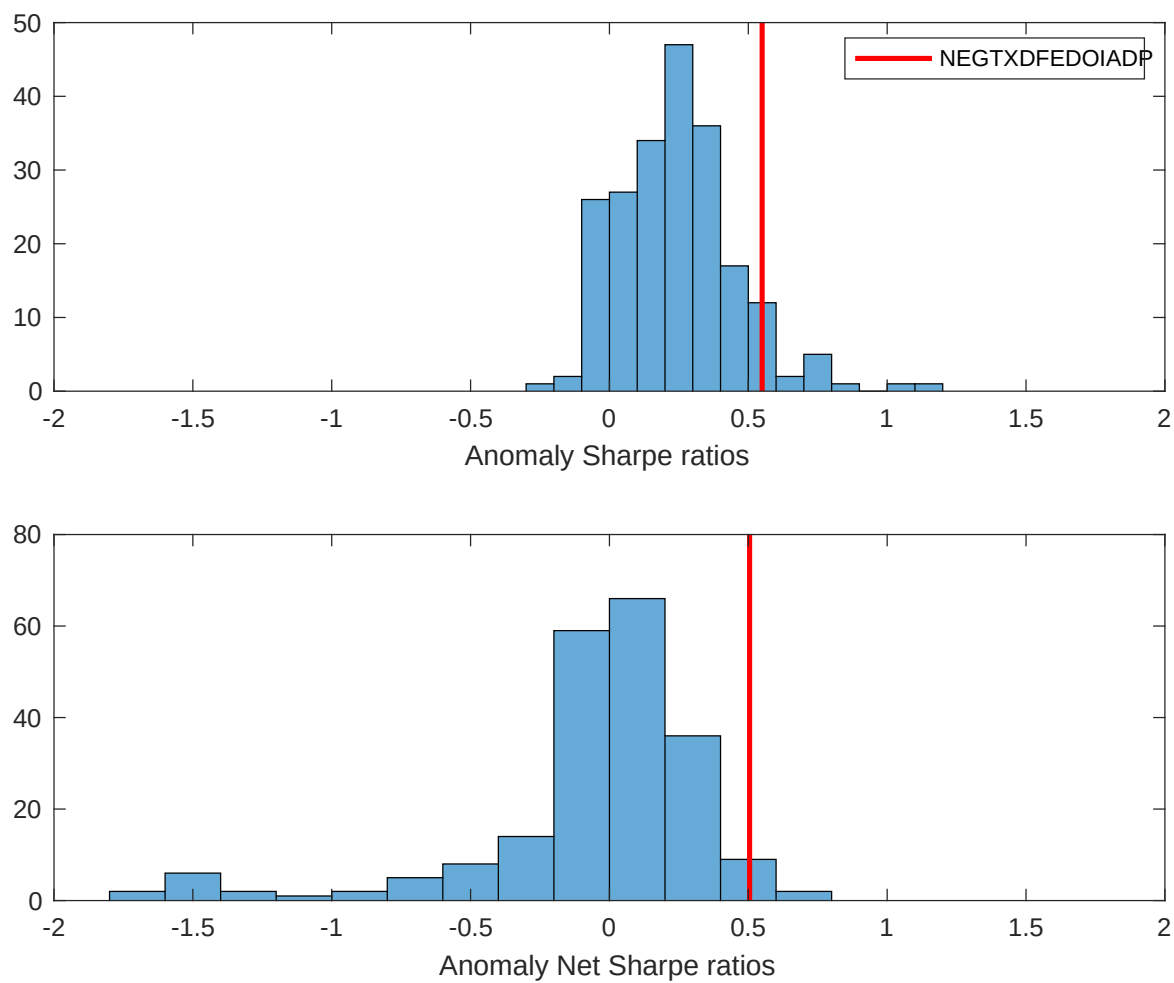


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

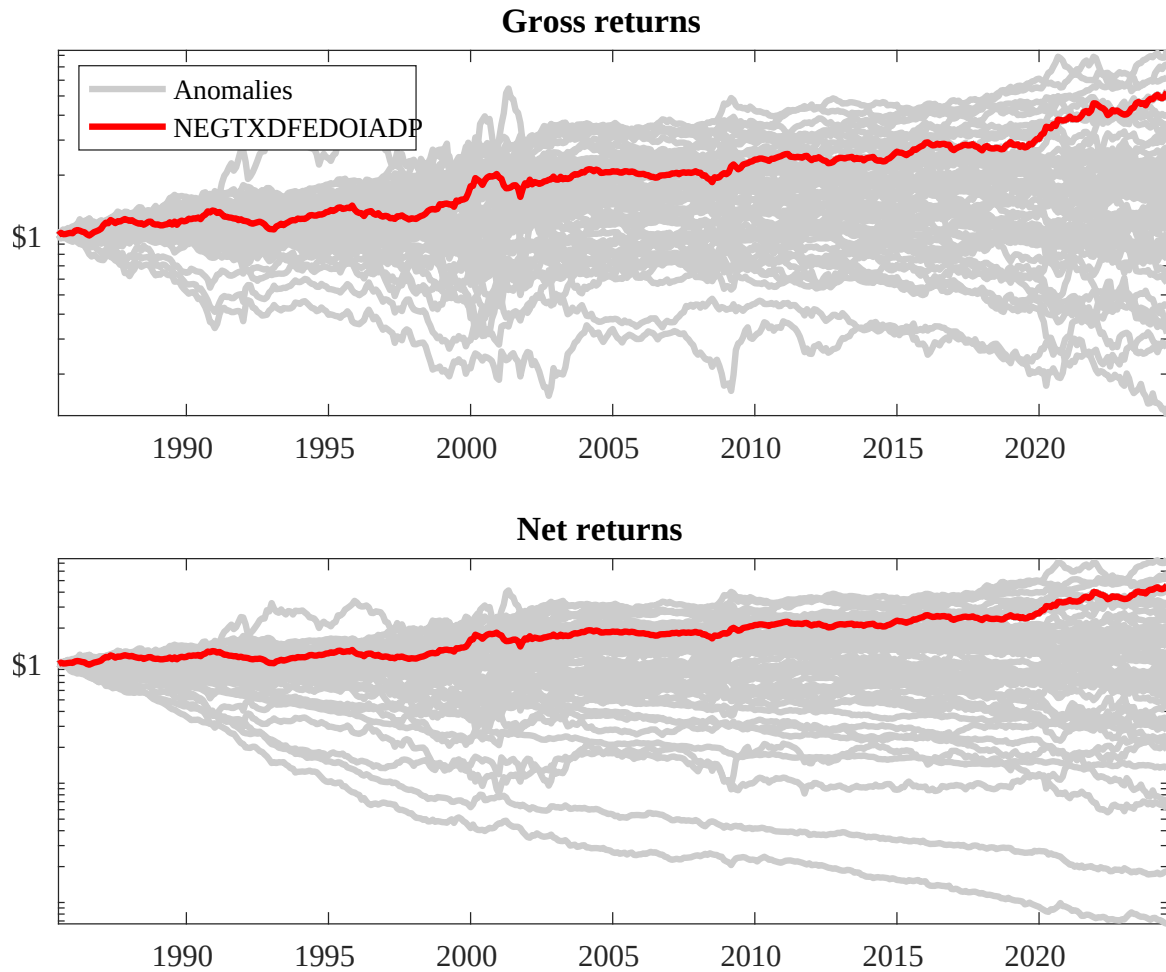
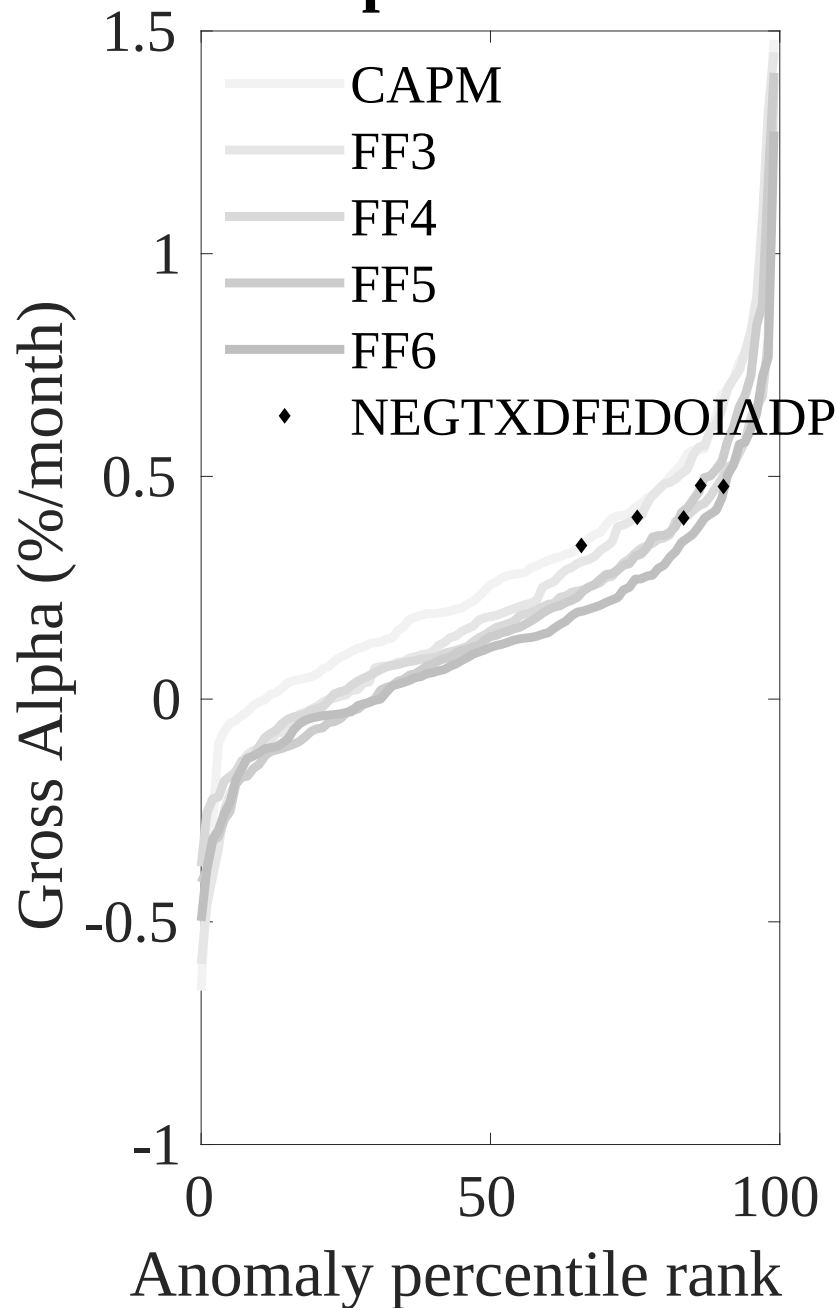


Figure 3: Dollar invested.

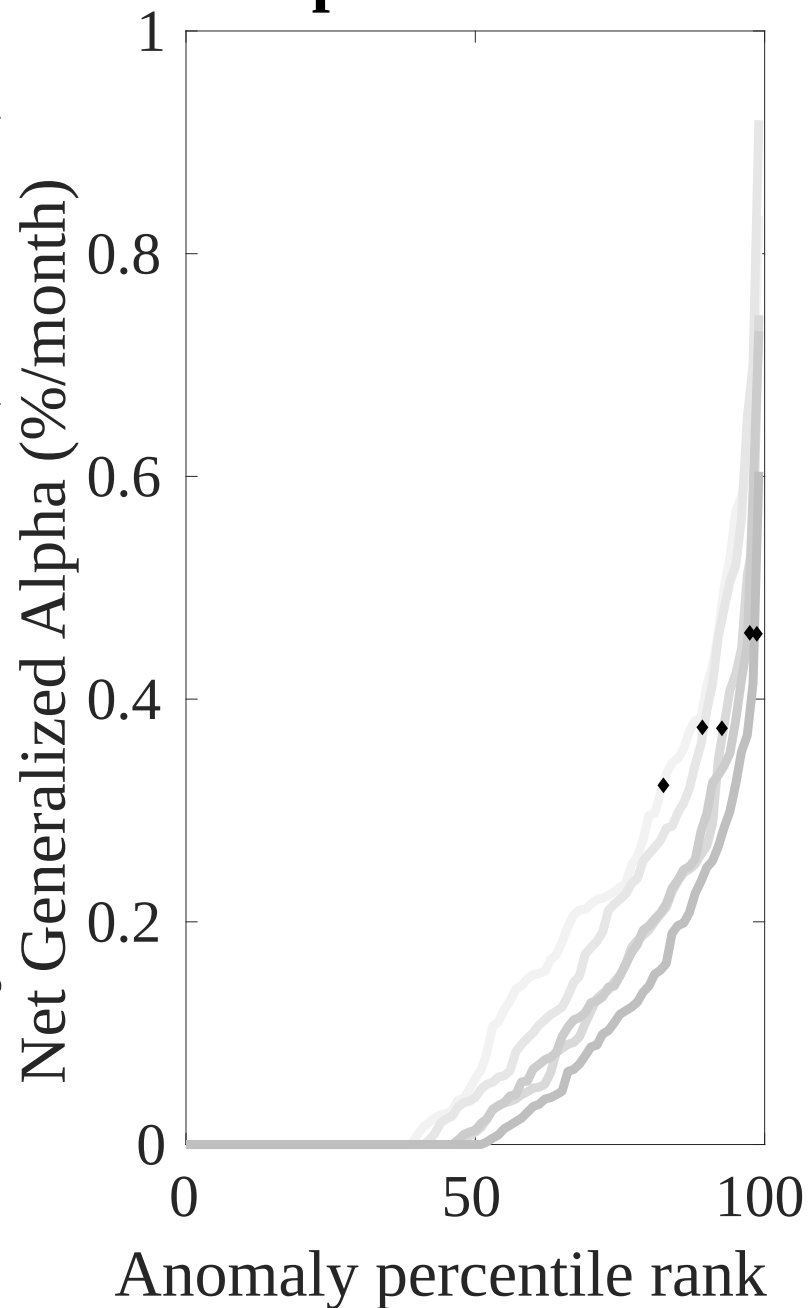
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



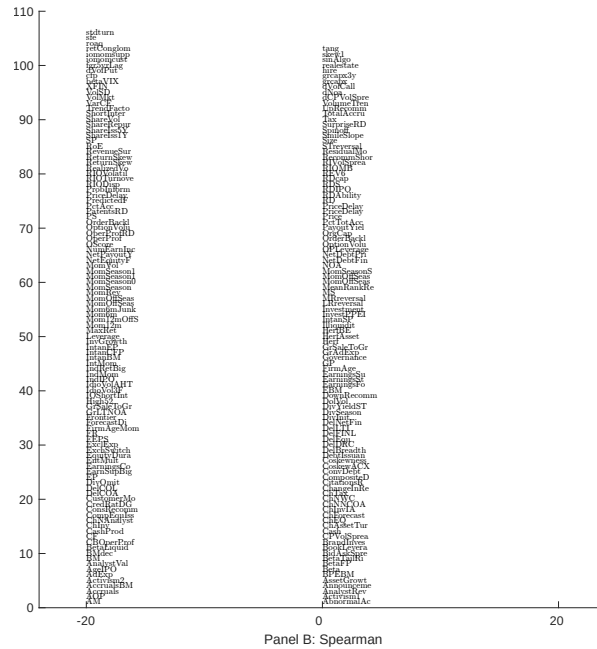
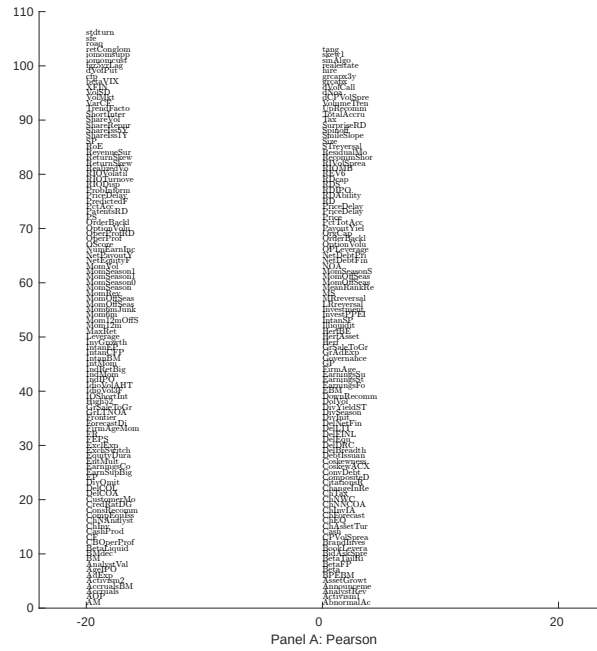


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

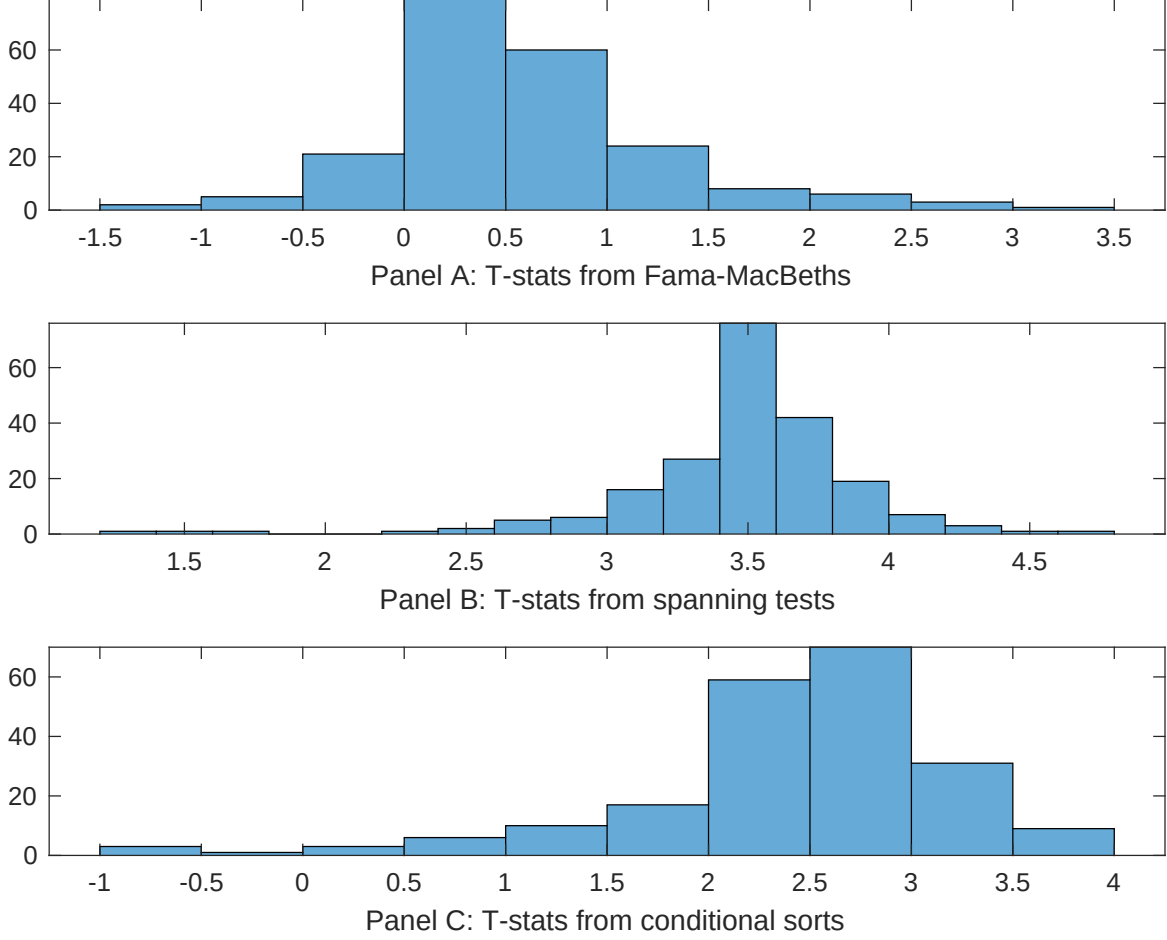


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10 [3.69]	0.16 [6.32]	0.10 [3.50]	0.93 [3.04]	0.10 [3.51]	0.10 [3.46]	0.12 [4.81]
TSI	0.78 [1.35]	0.18 [0.31]	0.24 [0.43]	0.34 [0.59]	0.39 [0.69]	0.34 [0.59]	0.43 [0.77]
Anomaly 1	0.62 [1.56]						0.11 [3.13]
Anomaly 2		0.25 [2.66]					0.23 [3.41]
Anomaly 3			0.18 [0.58]				-0.23 [-0.98]
Anomaly 4				0.27 [3.22]			0.20 [2.84]
Anomaly 5					0.13 [0.61]		-0.25 [-1.32]
Anomaly 6						0.85 [2.95]	0.79 [4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28 [2.84]	0.47 [4.63]	0.48 [4.73]	0.46 [4.56]	0.47 [4.56]	0.51 [5.00]	0.29 [2.90]
Anomaly 1	33.90 [8.54]						30.37 [7.01]
Anomaly 2		-13.35 [-2.49]					-1.87 [-0.29]
Anomaly 3			-8.48 [-2.40]				2.14 [0.48]
Anomaly 4				-25.89 [-3.87]			-9.98 [-1.45]
Anomaly 5					-12.58 [-3.00]		-6.98 [-1.62]
Anomaly 6						-13.55 [-3.62]	-3.78 [-0.87]
mkt	-6.58 [-2.77]	-0.34 [-0.14]	-0.21 [-0.09]	-1.30 [-0.54]	1.38 [0.56]	0.27 [0.11]	-5.33 [-2.14]
smb	-6.87 [-2.00]	-5.10 [-1.33]	-7.16 [-1.95]	-0.88 [-0.22]	-7.09 [-1.94]	-5.87 [-1.60]	-3.35 [-0.86]
hml	-12.23 [-2.75]	-10.22 [-1.36]	-19.96 [-4.06]	1.74 [0.21]	-8.23 [-1.16]	-14.92 [-2.86]	9.45 [1.00]
rmw	-5.99 [-1.26]	-22.11 [-4.77]	-19.30 [-4.01]	-29.19 [-5.93]	-24.13 [-5.18]	-17.73 [-3.71]	-10.77 [-1.97]
cma	18.37 [2.92]	1.37 [0.20]	8.59 [1.29]	6.95 [1.07]	3.20 [0.49]	8.92 [1.36]	15.48 [2.28]
umd	0.42 [0.20]	0.48 [0.21]	-2.98 [-1.07]	1.58 [0.69]	-2.06 [-0.83]	-4.75 [-1.73]	-1.50 [-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

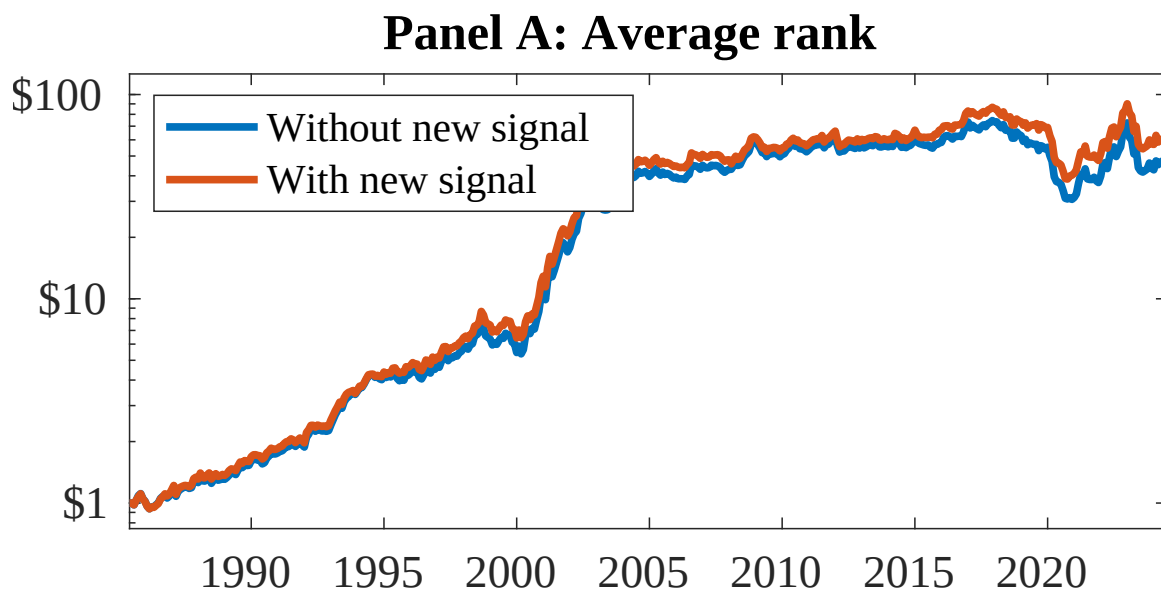


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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